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Averaging reduces variability...

We argued that bagging of trees will work because averaging reduces variability: if $U_1, \ldots, U_n$ are uncorrelated with variance σ^2, then

$$\mathrm{Var}[\bar{U}] = \frac{\sigma^2}{n}.$$

However, the B trees that are "averaged" in bagging are not uncorrelated! This will result into a smaller reduction of the variability.

Random forests have the flavour of bagging-of-trees, but they incorporate a modification that aims at decorrelating the trees.

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Random forests

Like bagging-of-trees, random forests predict via majority voting from classification trees trained on B bootstrap samples.

However, whenever a split is designed in each tree, the split is only allowed among $m(<< p)$ predictors randomly selected out of the p predictors.

The rationale: in a situation where there would be one strong predictor only, most bagged trees would use this predictor in the top split, which would result in highly correlated trees. The tweak above will prevent this, hence will lead to less correlated trees.

Random forests

Using $m = p$ would simply provide bagging-of-trees.

Using m small is appropriate when there are many correlated predictors.

Common practice:

- For classification (where majority voting is used), $m \approx \sqrt{p}$.
- For regression (where tree predictions are averaged), $m \approx p/3$.

In both cases, results are actually not very sensitive to m.

Random forests

Random forests

- are nonparametric and efficient,
- can deal with a large number of predictors (high dimensions ok),
- can cope with both small and large sample sizes (Big Data ok),

but they

- rely on a rather black box model, and
- are not supported by strong theoretical results.

Let us look at efficiency...

A simulation

We repeated $M = 1000$ times the following experiment:

(1) Split the `channing` data set into a training set (of size 300) and a test set (of size 162);

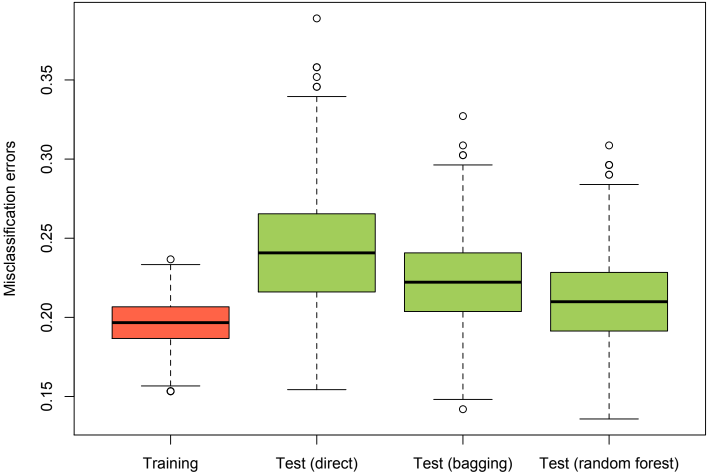
(2a) train a classification tree on the training set and evaluate its test error (i.e., misclassification rate) on the test set;

(2b) do the same with a bagging classifier using $B = 500$ trees;

(2c) do the same with a random forest classifier using the randomForest function in R with default parameters ($B = 500$ trees, $m \approx \sqrt{p}$).

This provided $M = 1000$ test errors for the direct (single-tree) approach, $M = 1000$ test errors for the bagging approach, and $M = 1000$ test errors for the random forest approach.

A simulation



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Measuring importance of each predictor

Because bagging-of-trees and random forests are poorly interpretable compared to classification trees, the following is useful.

The importance, v_j say, of the jth predictor is measured as follows:

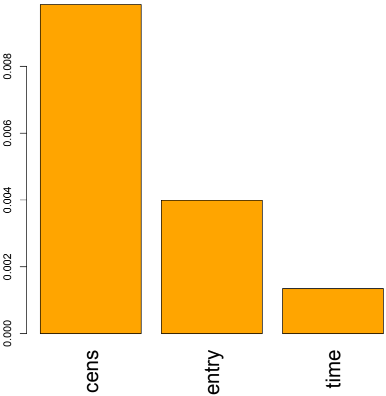
For each tree (i.e., for any $b = 1, \ldots, B$),

- the prediction error on OOB observations is recorded, and
- the same is done after permuting randomly all values of the jth predictor (which essentially turns this predictor into noise),

the difference between both errors is then averaged over $b = 1, \ldots, B$ (and normalized by the standard error—if it is positive), yielding v_j.

(A similar measure is used for regression, based on MSEs).

Measuring importance of each predictor



Importance of each predictor (decreasing order)